

Ramprasad Subramanian

Surebuckle71@proton.me | +49 176 35110221 | Berlin, Germany
linkedin.com/in/ramprasadsbramanian | github.com/Surebuckle71

Summary

M.Sc. Data Science student with an applied mathematics background. Takes work end to end, from experimental design and statistical analysis to tested, deployed model APIs, across NLP and LLMs, causal inference, and forecasting.

Education

University of Europe for Applied Sciences, Potsdam | *M.Sc. Data Science* *Thesis: September 2026*
Thesis: LLM-Guided Socratic Hint Generation with Student-State Awareness for Intelligent Tutoring Systems

Flame University, Pune, India | *B.Sc. Applied Mathematics* *May 2023*

Skills

Languages: Python, R, SQL

Statistics & Experimentation: Hypothesis testing, A/B testing, power analysis, multiple-comparison correction, bootstrap confidence intervals, causal inference (difference-in-differences, propensity scores, CATE meta-learners)

Machine Learning: Classification & regression, feature engineering, class-imbalance handling (SMOTE), model calibration, cost-sensitive thresholding, time-series forecasting with walk-forward validation, SHAP explainability

NLP & LLMs: Transformer fine-tuning (BERT, RoBERTa, XLM-RoBERTa), QLoRA, prompt engineering, LLM evaluation

Tools: pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers, Prophet, FastAPI, Docker, GitHub Actions (CI/CD), Git, tidyverse/ggplot2, Tableau

Projects

M.Sc. Thesis: LLM-Guided Socratic Hint Generation | *Python, PyTorch, Hugging Face, Unsloth, QLoRA* *2026*

- Fine-tuned Qwen2.5-3B-Instruct with QLoRA on MathDial tutoring dialogues within a 6 GB VRAM budget to generate Socratic hints conditioned on a student-mastery estimate inferred from the dialogue.
- Evaluated five conditions on 241 held-out dialogues with LLM-judged hint quality: fine-tuning improved quality over the prompted base model; state-conditioning produced no significant gain, and a small first-run deficit did not replicate over three seeds.
- Built a same-history counterfactual diagnostic with a placebo check, showing that most of the model's response to an injected state came from the wording of the state line.

Transformer-Based Aspect Sentiment Analysis | *Python, PyTorch, Hugging Face Transformers, FastAPI*

- Fine-tuned BERT, RoBERTa, and XLM-RoBERTa and compared them with LSTM and TF-IDF + SVM baselines on 1.31M Zomato review instances.
- Found that 97.5% of test reviews duplicated training text, inflating baseline accuracy (SVM 97.0%, LSTM 98.3%); on the 4,947 unseen reviews, RoBERTa generalized best (83.0%), reversing the apparent model ranking.
- Designed a distant-supervision evaluation for per-aspect sentiment without gold labels (83.7-90.1% accuracy across four aspects); shipped the model as a tested FastAPI /predict service with CI.

Telco Churn Risk API | *Python, scikit-learn, SHAP, FastAPI, Docker, GitHub Actions*

- Compared models on ~7,000 telecom customers; logistic regression beat random forest and gradient boosting in 5-fold CV (ROC-AUC 0.845 vs. 0.819 and 0.833), then tuned and calibrated it.
- Set the decision threshold (0.10) from assumed retention costs (\$50 per offer, \$500 per missed churner), for an estimated \$106,930 saved per 1,000 customers scored versus flagging no one.
- Deployed a Dockerized FastAPI service with /predict and SHAP-based /explain endpoints (p95 latency 17 ms, 94% test coverage, CI on GitHub Actions).

Mobile Game A/B Test Analysis (Cookie Cats) | *Python, Jupyter*

- Analyzed a ~90,000-player test of paywall gate placement: sample-ratio check, power analysis (0.93-point minimum detectable effect at 80% power), two-proportion z-tests with bootstrap CIs, and Benjamini-Hochberg correction.
- Recommended keeping the gate at level 30 after finding significantly higher day-7 retention (corrected $p = 0.0047$); simulated how peeking at six interim checks inflates the false-positive rate to 20%.

More projects (github.com/Surebuckle71): CATE estimation with S/T/X/DR-learners (X- and DR-learners within ~3% of oracle targeting); student-dropout prediction across 10 models (F1 89.2%, ROC-AUC 96.3%, FastAPI + CI); LaLonde causal benchmark; staggered difference-in-differences on state minimum wages (Callaway-Sant'Anna); CAISO day-ahead demand forecast (MAPE 2.89% vs. 8.04% baseline).

Experience

Corizo | *Data Science Intern, Remote* *June 2022 - Aug 2022*

- Built and evaluated predictive models in Python across 3+ projects, including a wine-cultivar classifier (97% test accuracy).

Volunteering & Certifications

Aasraa Trust | *Mathematics Curriculum Volunteer, Remote* *July - Aug 2021*

Designed math worksheets and learning materials for 50+ underprivileged schoolchildren across multiple grade levels.

AWS Academy Graduate - Cloud Foundations, AWS Academy, June 2025